Mapping the Shadow Economy

AI-Driven Identification of Informal Markets in Brazil

Carolina Abdalla Braga

The University of Chicago
Master of Public Policy

February 2026

Identification Strategy: The Shadow Logic

This research identifies economic activity that is physically present but infrastructurally "invisible" to formal sensors.

- **Detection:** Street-level imagery identifies micro-clusters of vendors via Computer Vision.
- **Formal vs. Informal Proxy:**
  - **Formal:** High Vendor Count + High Nighttime Radiance.
  - **Informal:** High Vendor Count + Low Nighttime Radiance.
- **References:** Extending Henderson (2012) and Chen & Zhang (2021), who famously use satellite nighttime lights to proxy formal GDP, this study inverses their logic to map "dark", unrecorded economic growth.

Methodology: Datasets & Technical Stack

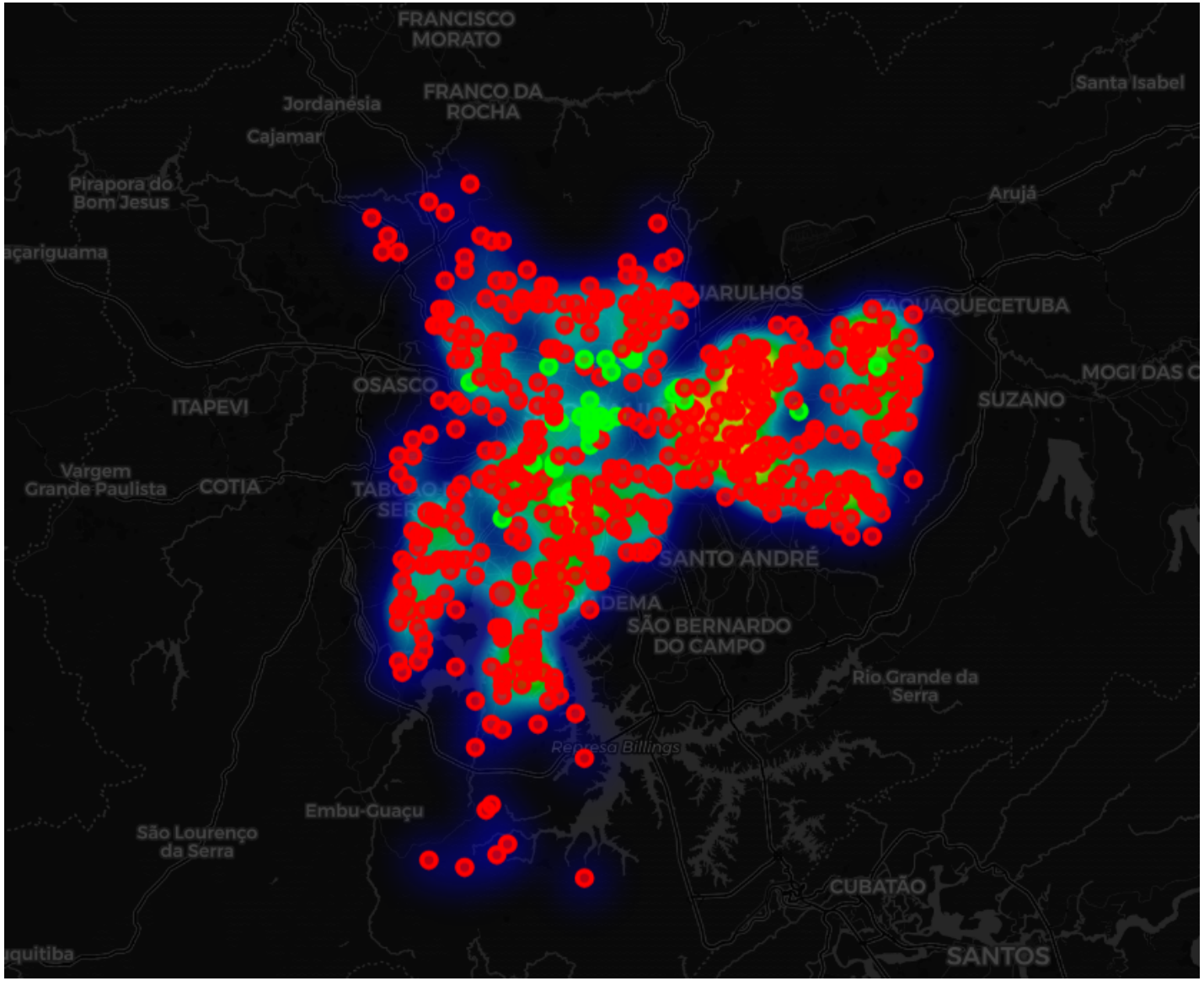
To process the urban grid of São Paulo, I built a fully automated Python spatial pipeline.

Core Datasets & Tools:

- **Street Imagery:** Google Maps Static API (scanned via official IBGE municipal shapefiles).
- **Computer Vision:** Roboflow Platform & custom-trained **YOLOv11** model.
- **Satellite Data:** VIIRS Nighttime Lights extracted via Python's `rasterio` engine.
- **Socio-Economic Data:** IBGE PNAD-C Household Survey.

Figure: YOLOv11 model detecting informal vendor structures.

Visualizing the Shadow Economy



Heatmap of Informal Commercial Density

Data Construction: The Engine Behind the Map

To capture "markup" variance across São Paulo, a **Wealth Index (Ω)** adjusts the **PNAD-C household survey's** baseline income based on commercial geography (e.g., Tier 1 Elite = 1.8x multiplier; Tier 5 Periphery = 0.75x).

These datasets feed into two primary econometric proxies:

1. Revenue Estimation

$$Revenue = Activity \times 1.5_{workers} \times (PNAD_{income} \times \Omega_{district})$$

2. The Shadow Index (S_i)

$$S_i = \frac{Total_Activity}{\ln(1 + VIIRS_Radiance)}$$

Note: The Shadow Index penalizes well-lit areas, isolating high-density markets functioning without formal infrastructure.

Empirical Results & Policy Impact

The Shadow Index (S_i) successfully isolates "Giant" informal hubs from wealthy formal districts.

District	Wealth Index	Avg. Radiance	Informal Revenue	Shadow Index
Pinheiros	1.80	112.45	R$ 12.3M	12.40
Brás	1.50	42.15	R$ 88.6M	**76.22**
Santo Amaro	1.20	65.30	R$ 15.1M	24.15
Lajeado	0.75	8.12	R$ 9.4M	**45.80**

- **The Shadow Giant:** Brás generates 7x more informal revenue than wealthy districts like Pinheiros, despite 60% less light.
- **Policy Implication:** Provides a high-resolution, targeted database for urban planning and social protection resource allocation.

Key References

- Chen, M., & Zhang, S. (2021). Measuring the regional non-observed economy in China with nighttime lights. *IJOEM*.
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